HEALTH INSURANCE ANALYSIS REPORT: STEPS AND CHALLENGES

Introduction

This report details the implementation of the **UrbanNest Health Analytics** project, which focuses on predicting medical insurance charges using patient demographic and behavioral data. The primary goal of this analysis is to apply machine learning—specifically Multiple Linear Regression—to accurately predict annual healthcare costs and evaluate how features such as age, BMI, and smoking habits impact financial risk. The following sections outline our analytical approach, data encoding strategies, predictive modelling outcomes, and the specific challenges encountered during execution.

First Phase: Data Integrity Checks

Healthcare Insurance Charge Prediction Capstone

Dataset: Medical Cost Personal Dataset (insurance.csv)

The objectives

In this stage, we were starting from the raw dataset and learning about the information present in the data before proceeding to building any machine-learning model.

We:

1. Load and inspect the dataset.
2. Understand the rows, columns, data types, and variables.
3. Check for missing values.
4. Check for duplicate records.
5. Validate basic value ranges and categories.
6. Explore numerical and categorical variables.
7. Investigate relationships with the target variable, charges.
8. Identify unusual observations without automatically deleting legitimate healthcare cases.
9. Create a clean working dataset for the remaining EDA.

Importing the Python libraries needed.

The following are the libraries we are going to need to achieve our stage 1 workflow:

- **pandas**: to load, inspect, clean, and summarize the data.
- **numpy**: for basic numerical operations.
- **matplotlib**: to create appealing visualizations.
- **seaborn**: to create appealing and clearer statistical plots.

Loading the dataset

We began with the original insurance CSV file exactly as supplied.

At this stage, **we did not clean or transform anything yet**. The purpose was to inspect the raw data first and draw insight from it in order to understand it.

Checking the data types and general structure of the data

we wanted to make sure that:

- numerical variables are stored as numbers;
- categorical variables are stored as text;
- the target variables are numerical because this is a regression problem.

what we understand each variable represent:

Column	Meaning	Type	Role
age	Patient age	Numerical	Predictor
sex	Patient sex	Categorical	Predictor
bmi	Body Mass Index	Numerical	Predictor
children	Number of children/dependents	Numerical count	Predictor
smoker	Smoking status	Categorical	Predictor
region	Residential region	Categorical	Predictor
charges	Individual medical insurance charge	Numerical	**Target**

We want to check for missing values in our raw dataset

This was done because missing values can cause errors or misleading analysis, so we check them before doing any EDA.

We calculated the number of missing values in each column.

Checking for duplicate records in our dataset

After checking for missing values, in order to make sure that our dataset is clean, we also check for duplicates records A duplicate record means that a row appeared more than once**,**

This was checked separately from missing values.

Data Cleaning

After checking the data integrity, we discovered that there was a duplicate row in our dataset so we had to remove the duplicate to prevent any future errors with our analysis and to have the best data for our model to use for prediction.

Then we added helper columns like BMI category and Age group to enable easier grouping during our analysis.

Exploratory Data Analysis Section

Now that we have passed data integrity checks, we moved into the EDA (exploratory data analysis).

For each major chart, we are asking:

1. What pattern do we see?
2. What could the pattern mean in the healthcare/insurance context?
3. What should we avoid claiming from the pattern?
4. what deductions or information is the pattern telling us?
5. what is the current situation of the insurance company based on the pattern from the data?

We performed univariate, bivariate, and multivariate analysis to figure out the patterns, insights and stories in the dataset.

We used visuals like histogram, column chart, box plot, heatmap, scatter plot to figure out the hidden details in the dataset.

During our exploratory data analysis, we examined feature distributions and relationships to uncover the primary drivers of healthcare costs. Our initial analysis of the target variable revealed that medical charges follow a heavily right-skewed distribution, where the majority of policyholders incur relatively low expenses, while a small subset of high-cost patients drives up the overall average.

Second Phase: Feature Engineering & Linear Regression Modelling

Now that we have completed data cleaning, EDA, and SQL analysis, we move into the core of the project: building a Linear Regression model that predicts insurance charges based on patient attributes.

Why We Do This

The insurance company currently prices policies using rough assumptions. Our goal is to replace those assumptions with a **data-driven model** that learns the relationship between patient features (age, BMI, smoking status, etc.) and their medical charges.

Our Roadmap

1. Feature Engineering — Convert categorical text into numbers the model can understand.
2. Separate Features (X) from Target (y) — Distinguish what we predict from what we predict *with*.
3. Train/Test Split — Reserve 20% of the data to fairly evaluate the model.
4. Train Linear Regression — Fit the model using Scikit-learn.
5. Evaluate & Interpret — Measure performance and translate results into business insights.

To build our predictive model, we utilized Multiple Linear Regression to establish a quantitative relationship between policyholder characteristics and their medical charges. After processing the features using One-Hot Encoding to handle categorical variables without introducing multicollinearity, we split the dataset into an 80% training set to fit the model and a 20% testing set for evaluation. The resulting linear regression equation achieved an R^2 score of approximately 80% on the unseen test data, explaining the vast majority of variance in healthcare costs with a root mean squared error (RMSE) of around $5,956. Examining the learned coefficients confirmed that smoking status was by far the strongest financial predictor, adding approximately $23,080 to baseline charges, followed by steady incremental increases for each additional unit of BMI and year of age. Finally, we validated the model using residual diagnostic plots—including actual versus predicted comparisons, residual distribution checks, and Q-Q plots—to assess predictive performance and verify regression assumptions.

CHALLENGES

One of the challenges we faced was during the encoding of the independent variables, since we added helper columns like BMI category and Age group, we had to find a way to encode only the needed columns which is region, sex, smoker.

Collaborating. finding the best method in which everyone could participate in the project was a difficult task since we had to make sure everyone was aware of what we are doing. Each person

had to attempt each task individually then combine the results into one big work, combining results from each person into one big project was not an easy task.

Writing reports and documenting our progress, findings and recommendations was also a challenge since we are amateurs in these aspects. It was difficult finding the right words to express ourselves and communicate our findings.